

Table 1. Specification of the test and relevant samples

No. of blind holes	1	2	3
Thickness of bottom of blind hole (mm)	0.642	1.361	2.258
Speed of flyer (m/s)		172	
Tensile stress (MPa)		1470	
Duration of tensile stress (μ s)	0.26	0.50	0.80
Number of microcracks (no./mm ²)	324	454	722

polished with an Abramin automatic grinding and polishing machine. Then, with a Quantum 520 image analysis system, damaged regions on the surfaces were examined. For each sample, microcracks in about 10–20 viewfields of 0.025 mm² each were counted and summed to reduce fluctuations resulting from mesoscopic inhomogeneities. Figure 3 shows the design and dimensions of the flyer used in the case study. Table 1 lists specifications of the test and relevant samples. Figure 4 shows the sequential pictures of microcracks on sectional surfaces of samples ($<0.1 \mu$ s, 0.26μ s, 0.5μ s and 0.8μ s, respectively).

To estimate possible errors involved in measurements of microcracks in the test, some preliminary measurements were made. Firstly, the transverse isotropy and homogeneity of the aluminium alloy plate were examined. On the plate surface grains were found to be roughly equiaxial, with ellipticity (~ 0.91). Whilst the ellipticity of grains on two sectioned surfaces perpendicular to each other are both 1.8 (Han, 1993). Then, from an arbitrary point on the plate surface we drew a straight line and counted the number of grains the line passed. The measurements showed that the grain number is independent of orientations of the straight line with variation less than 5% (Ling and Bai, 1994). These results justify statistical equivalence of measurements on different sectioned surfaces.

Then, possible sources of scatter and errors were estimated. The scatter of numbers of second phase particles measured on different sectional surfaces is about 5.2% (Han, 1993), whereas numbers of microcracks in tested samples of different shots with similar loading conditions have greater variation, up to 10–20%, even to 40%. Nevertheless, statistical features of the obtained data are similar (Luo, 1988). Finally, the variation from sectional to volumetric transformation was examined numerically. By taking parameters similar to our tests i.e. the total microdamage on the sectional surface is $10^3/\text{mm}^2$ and the typical size of microdamage is $10 \mu\text{m}$, the variation involved in the transformation (25) is 3% (Bai *et al.*, 1993). Thus, we thought that our experimental data were representative.

Before going further, we should also examine the scatter of obtained data in the case study. For each sample, e.g. that of 0.2μ s, we have $36 \times 24,663 \mu\text{m}^2 \approx 0.88 \text{ mm}^2$ viewfields. The deduced rate of nucleation are also listed in Table 1. The least square fitting gives the nucleation rate of $928/\text{mm}^2 \mu$ s. So, except for an overshoot at the beginning, the nucleation rate of microcracks seems to be time-independent in the time range up to 0.8μ s. This qualitatively agrees with simulations, in which just before EIC, only very few coalescences occur. Therefore, we thought that the model of ideal microcracks, i.e. negligible coalescence and annihilation, works in the case study.

Figure 5 shows the sectional number density distributions of microcracks $m(t, c', \sigma_0)$ appearing at different time intervals. The Fourier approximation technique is applied to eliminate fluctuations. Figure 6 shows the smoothed number density distributions $m(t, c', \sigma_0)$ and nucleation rate $m_N(c', \sigma_0)$. Figure 7 shows the volumetric distribution $n(t, c, \sigma_0)$ and $n_N(c_0, \sigma_0)$ completed in accord with the equation of transformation and its solution (24) and (25). Figure 8 demonstrates that the average growth rate of microcracks A is dependent upon the size c as well as time t derived from eqn (10). This is not physically meaningful. Figure 9 shows the growth rate of microcracks V versus the size c and the nucleation size c_0 in our experimental range from 0.26μ s to 0.80μ s. Figure 10 provides the growth rate of microcracks V as a function of c_0/c in the same range of time, but unfortunately it was found that unlike the expressions derived by Mott (1948), Berry (1974) and others, the physical growth rate V is not the unique fluctuation of c_0/c . Finally Fig. 11 shows that the data of growth rate V collapse to a function of a single variable—incremental